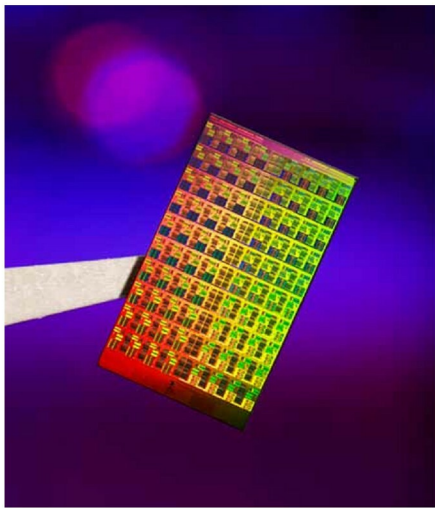


SEMICONDUCTOR FABRICATION

Innovation Continues, But Goals Change

Researchers are experimenting with carbon nanotubes to phenomenally change the way semiconductor devices are made. At the same time, there are innovations to address immediate needs such as improving yield, decreasing time-to-market, reducing leakage power, thermal management in multi-die stacks, new layout requirements and so on



■ JANANI GOPALAKRISHNAN VIKRAM

The 'chips' are everywhere today, right from mobile phones and computers to microwave ovens and washing machines, and even in children's toys. There are millions of chips in the world and more are being produced every day, but did you know that these are still produced by a countable number of manufacturers across the world? A fab, or a semiconductor manufacturing plant, is a technology- and resource-intensive factory that could cost anywhere between 1 billion and 10 billion dollars, or even more to set up.

The challenges and resulting improvements in semiconductor fabrication technology can be attributed to the complexity and sensitivity of the whole process. It could take six to eight weeks per batch right from wafer processing and die preparation to integrated circuit packaging and testing. What's more, it is a very sensitive process too. As chips get smaller, purer materials are sought. The 'clean rooms' of fabs, where most of the processes happen, need to be cleaner than you